«LangGraph子图组装者»
CalendarReadWorkflow

-CalendarAgent calendarAgent
-ToolExecutionService toolExecution
-ConversationArtifactService artifactService

+build() : CompiledStateGraph
-plan_calendar_read(state)
-execute_calendar_plan(state)

创建、校验并写入State

execute_read_tool()

提取可信摘要

提取卡片字段

persist()

«领域规划器»
CalendarAgent

+plan_read(userContent) : ToolInvocation?

«Pydantic计划契约»
CalendarReadPlan

+action list_events|find_free_time
+tool_name string
+arguments dict

+from_invocation(invocation) : CalendarReadPlan
+to_invocation() : ToolInvocation

«任务、审计与进度编排»
ToolExecutionService

+execute_read_tool(run, context, invocation) : ReadToolExecutionResult
+invoke_read_tool(run, context, invocation) : string

«安全结果摘要»
ToolResultSummary

+status succeeded|failed
+tool_name string
+summary dict

«前端卡片数据»
ConversationArtifact

+artifact_type calendar_events|calendar_free_time
+payload dict

ConversationArtifactService

+persist(run, artifacts)

提出候选调用

校验后重建调用

prepare() + invoke()

返回执行结果

发布工具进度

artifacts_ready

ToolInvocation

+tool_name string
+arguments dict

«工具安全入口与分发器»
ToolGateway

+prepare(...) : PreparedToolCall
+invoke(prepared) : ToolResult
+get_definition(toolName) : ToolDefinition

ReadToolExecutionResult

+status succeeded|failed
+tool_context string
+output_summary dict

«Redis事件发布器»
RunEventPublisher

+publish_after_commit(event)

ConversationArtifactRepository

+upsert_for_run(...)

工具审计、步骤、事件

查询已注册工具

校验连接、scope、租户

保存Artifact

«已注册工具表»
ToolRegistry

+get_handler(toolName) : ToolHandler

ProviderConnectionRepository

+find_active_connection(...)

«数据库»
PostgreSQL

tool_calls
run_steps
run_events
conversation_message_artifacts

按tool_name找到Handler

«日历工具适配器»
CalendarMCPToolHandler

+invoke(context, arguments) : ToolResult

call()

«MCP客户端»
CalendarMCPClient

+call(toolName, context, arguments) : ToolResult

MCP请求

«外部工具服务»
CalendarMCPServer

+calendar.list_events()
+calendar.find_free_time()